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SUMMARY

Computer Science undergraduate aiming at research-oriented AI engineering. Foundation so far: a self-written Python and SQL application, a four-person full-stack web project, and coursework in algorithms, discrete mathematics, probability, object-oriented programming and databases. Currently strengthening linear algebra, probability and optimisation so that I can implement learning algorithms from first principles; strong interest in deep learning, reinforcement learning and LLM post-training.

RESEARCH INTERESTS

Deep learning optimisation • reinforcement learning and post-training (reward modelling, RLHF, DPO, GRPO) • LLM and VLM architectures, multimodal learning • efficiency topics: Mixture-of-Experts, state-space models. *(Study interests; not yet hands-on.)*

EDUCATION

[B.Tech / B.E.] in Computer Science and Engineering — [University Name] *[Start Year] – [End Year]*

[City], India • Current year: [e.g. First Year] • Expected graduation: [Month Year] • [CGPA / percentage, optional]

Senior Secondary (Class XII), CBSE — Delhi Public School, Pune *2024 – 2025*

Computer Science project track; the Python project below was written for this course

Coursework and current study: Data Structures & Algorithms; Discrete Mathematics and Probability; Object-Oriented Programming; Database Management Systems and relational SQL; [add others, e.g. Linear Algebra, Computer Organisation].

TECHNICAL SKILLS

- **Programming:** Python (project experience); C and C++ (coursework); SQL (joins, subqueries, aggregate functions, grouping, filtering).
- **Web and backend:** HTML and CSS (fundamentals); JavaScript (basics); Node.js and Express (recently started); React/Next.js client work in a team project; consuming and debugging REST APIs.
- **Databases and tools:** MySQL; PostgreSQL with Prisma (schema, migrations, seeding); Git and GitHub; VS Code.
- **Currently learning:** NumPy (in progress) and introductory Pandas; Node.js backend development; linear algebra, probability and optimisation problem sets.

PROJECTS

Movie Streaming Platform *Python, MySQL*

Individual project • console application, single file, ≈820 lines • github.com/some1iu/movie-streaming-platform

- Wrote a menu-driven console application on top of MySQL Connector/Python that handles account creation and login, per-user genre preferences, movie search by title/genre/year, and rent or purchase transactions.
- Built a filter-driven SQL query builder that appends WHERE clauses only for the criteria the user supplies, using parameterised queries for the search path; modelled the data in three tables (user, movie, bill) and committed every transaction as a bill record.
- Implemented genre-based recommendations, retrieval of binary movie data from the database and itemised bill generation through file handling; centralised error handling for input, type, key and `mysql.connector` errors so the session exits gracefully instead of crashing.

ACM Incursion — Conflict-Aware Events & Expenses Manager *Team of four*

Next.js/React client, Express API, Prisma, PostgreSQL • github.com/chetisha28/incursion-acm

- Contributed to a full-stack platform for ACM student chapters that flags venue, time and audience collisions before an event is published, gates proposals behind peer endorsement, and tracks expenses against semester budgets.
- Worked on the Next.js App Router client and its REST integration, and contributed to parts of the Express/Prisma backend: JWT-protected endpoints, role-based access across viewer, executive and core tiers, Zod request validation and `Decimal(12,2)` money columns.
- Debugged integration problems across Prisma migrations, seed data and CORS configuration to get the client running against the API locally.

Open-Source Codebase Exploration — Typebot *Bun, Nx, Prisma, PostgreSQL*

Exploration only, no upstream contribution • github.com/some1iu/typebot.io

- Independently set up and explored a large self-hosted chatbot-builder monorepo: Bun and Nx workspace configuration, package and dependency graph, Prisma client generation and migrations, environment variables, and the PostgreSQL services required for local development; worked through dev-environment failures until the stack ran.

ACADEMIC FOUNDATIONS

- **Algorithms:** backtracking and recursion, working systematically through Sartaj Sahni's algorithms text; standard data-structure and complexity exercises.
- **Mathematics:** discrete mathematics and probability problem sets; coding-theory problems including binary symmetric channels and transmission probabilities.
- **Object-oriented programming:** compile-time (overloading) versus runtime (overriding) polymorphism; class design and inheritance exercises.
- **Databases:** relational schema design and SQL practice — joins, subqueries, grouping, aggregates and filtering.